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14. ABSTRACT
This TOP describes procedures for determining the center of gravity (CG) of wheeled vehicles. The CG provides information relative to roll stability, transportability, and input for mobility model programs.

15. SUBJECT TERMS

Center of Gravity	Suspension Method	Pendulum timing method
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US ARMY DEVELOPMENTAL TEST COMMAND
TEST OPERATIONS PROCEDURE

[1]Test Operations Procedure (TOP) 2-2-800 26 September 2006
DTIC AD No. ******

CENTER OF GRAVITY

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1. SCOPE.

This document describes procedures for determining the location of the Center of Gravity (CG) of wheeled and tracked vehicles. The CG provides information relative to roll stability, transportability, and input for mobility model programs. Multiple methods of measurement are described.

a. Center of gravity or center of mass is a primary system characteristic that impacts vehicle design and testing at virtually all levels. From basic system design, transportability studies to detailed vehicle dynamics analyses the location of a vehicle's center of mass must be accurately known. Modeling and simulation inputs always require center of gravity locations for sprung and unsprung masses. Measurement of the center of gravity can be made using many different methods. The selection of the appropriate test method can depend on many factors including the following:

(1) How the measurement will be used:
   a) Defining the test vehicle
   b) Simulating its dynamic performance
   c) Chassis development
(2) Size and form of the test item

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(3) Facility limitations
(4) Available instrumentation

b. All of the primary methods used to determine the center of gravity use principles of engineering statics and require accurate measurements of the vehicle and/or component weights and weight reactions as well as linear and angular measurements. Weight measurements are accomplished using either platform scales or calibrated load cells. The acceptable error of measurement is 0.5 percent of reading. The total vehicle weight should be made as a single measurement whenever practical. Individual wheel weight measurements produce errors that can be additive in nature and should be used with caution. Linear measurements should be made using steel tapes of appropriate length. The acceptable error of measurement is ±1/16- inch. Length measurements are often projected to the ground plane using plumb bobs or levels. Care should be taken with projected measurements to preserve the measurement accuracy of the linear measurement. Angular measurements are usually required only as a reference when lifting a vehicle for a weight reaction type measurement. The required angle is a relative difference of the at-rest vehicle and the intermediate measurement condition. One should strive for chassis attitude angles of 15- to 25-degrees on the intermediate lifts to minimize the overall errors in longitudinal and height dimensions of the center of gravity location. The combined error increases dramatically as the included angle between the two intermediate lifts decreases. As an example, the longitudinal and vertical location of the center of gravity of a 15,000 lb vehicle changes from ±4mm (±0.15 inch) for an included angle of 50 degrees to±10 mm (±0.40 inch) at 30 degrees. Final parameter accuracy will require calculation for each test as it will be based on actual measured weights and linear dimensions. The highest angular lift should be conducted based on vehicle interference with the ground plane and/or lifting (crane) limits.

c. Reduction in random errors produces an improvement of the actual experimental method by continual refinement of the techniques required to conduct the test. Some of the common sources of random errors occur from the following.

(1) Suspension compression and/or droop
(2) Ride height changes
(3) Excessive tire deflection
(4) Sloshing/movement of fluid payloads in fuel tanks and reservoirs
(5) Track and suspension sag
(6) Changes in the measured total vehicle weight

d. The suspension method is based on the fact that a vertical line through the point of suspension will pass through the mass center of a freely suspended body. The reaction method and the weight method are based on the fact that when a body is in static equilibrium, the sum of the moments about an axis of rotation is zero. The pendulum timing method (also used for determination of vehicle moments of inertia[1, 2**] is based on the fact that the CG location can be calculated using the periods of oscillation measured when the test item is suspended and swung in a pendular motion at two different pendulum lengths.

e. Also included is a procedure for calculating the combined CG location of two or more

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masses when attached to each other and considered a single unit.

f. International Test Operating Procedure (ITOP 2-2-800(1)) FR/GE/UK/US Tracked Vehicle Center of Gravity should be consulted when a tracked vehicle is being considered. Facilities, instrumentation, test methods, and data presentation share many similarities with the wheeled vehicle procedures.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities.

Item	Requirement
Crane	To suspend or lift vehicle.
Transit or theodolite	To project a vertical line.
Vertical steps	To raise front or rear of vehicle.
Bed plate	Level (±0.1%) surface of sufficient area to support the test item.
Rotating hubs with pendulum arms (pendulum timing method)	To swing vehicle platform.

2.2 Instrumentation.

Devices for Measuring	Measurement Accuracy
Weight	±0.5% of reading
Length	±1/16 inch
Angle	±0.1 degree
Time	±0.001 second

In general, accuracy reflecting the use of available, cost-effective transducers and good mechanical measurements practice exceeds the specified measurement accuracies. English units of measurement are used for this test method as the test center's weight and linear measurement devices are calibrated in English units. Final conversion to SI units can be accomplished using the following conversion factors:

Multiply	By	To obtain
Inches	25.4	millimeters
Pounds (mass)	0.4534	kilograms
Pounds (force)	4.448	newtons

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3. REOUIRED TEST CONDITIONS.

Ensure that:

a. Test vehicle is in the desired load configuration – full payload condition (real or simulated payload) is mandatory; other load conditions (e.g., curb weight) are as required.

b. Specific vehicle configuration standards are defined in military[3] and Federal[4] standards. The following military standards are presented below.

1) The curb weight (CW) is defined as the empty vehicle weight, full compliment of fuels, fluids, lubricants, coolant, integral survivability and Basic Issue Items (BII).

2) The gross vehicle weight (GVW) is defined as the curb weight plus the weight of the crew and their individual equipment and weapons, trailer tongue weight, supplemental armor, and payload (payload includes mission essential support equipment, unique digital appliqué, and non-mission essential kits, such as cargo compartment heater kits and cargo compartment protection kit). The planning factor per soldier with individual equipment is 343 pounds.

3) The gross combination weight (GCW) is defined as the gross vehicle weight plus the combined weight of the towed load.

4) The gross combat loaded vehicle weight includes vehicle, fuel, lubricants, coolant, stowage, armor appliqué, ammunition and personnel. Combat weight and gross vehicle weight definitions are often interchanged.

5) Maximum shipping weight[5] is defined as the weight of the basic equipment plus the weight of any additional ancillary equipment, components, accessories, and cargo which are shipped with or contained within or attached to the equipment.

c. The Federal Motor Vehicle Safety Standard 571.3 “Definitions” provides specific definitions for the following commercial vehicle configurations.

1) Curb weight means the weight of a motor vehicle with standard equipment: maximum capacity of engine fuel, oil and coolant: and if so equipped, air conditioning and additional weight optional engine.

2) Gross vehicle weight rating or GVWR means the value specified by the manufacturer as the loaded weight of a single vehicle.

3) Gross combination weight rating or GCWR means the value specified by the manufacturer as the loaded weight of a combination vehicle.

4) Unloaded vehicle weight means the weight of a vehicle with the maximum capacity of all fluids necessary for operation of the vehicle, but without cargo, occupants, or accessories that are ordinarily removed from the vehicle when they are not in use.

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d. Suspension components are blocked or tied in their normal positions (with vehicle on level surface) before using any test procedure requiring suspension of part or the entire vehicle. This is required whenever stretching of springs or other suspension devices can possibly have an appreciable effect, due to weight shift, on the CG location.

e. Other pertinent vehicle configurations may be required based on the test plan or physical circumstance. Documentation of the vehicle configuration is critical. Written and photographic documentation are required.

f. Radiator, sump, hydraulic, and other reservoirs shall be filled to their specified working levels. The fuel tank(s) should be filled to capacity unless special conditions warrant an intermediate or empty load.

g. Tools, spare tire, and loose accessories and equipment, Basic Issue Items (BII) shall be complete as supplied and shall be in their normal stowage positions.

h. Tire pressures should be as specified in the manufacturer's operating instructions or, if a range of pressures are specified, the highest pressure should be selected to minimize tire deflection. The Tire and Rim Association's (TRA)[6] most current guidance should be consulted for appropriate tire pressure based on the tire dimensions, vertical loading and intended use. The military supplement of the TRA should be consulted for tactical truck applications[7].

i. Many of the measurements of linear dimensions required for calculation include averaging values on opposite sides of the vehicle. This reflects the existence of lateral and/or longitudinal asymmetries in most dimensions of a vehicle. These are generally small but can occasionally be significant. Thus, measuring values on both sides of the vehicle allow a check of asymmetry and averaging allows a truer representation of the vehicle dimensions. In certain instances the magnitude of the asymmetry may be desired (e.g. torsion bar offset, front and rear tread differences) and averaging may not be appropriate. In addition, side-to-side asymmetries can represent a specific design feature which should be documented[8].

j. Personnel engaged in rigging of suspension devices (cranes, cables, shackles, lift hooks, platforms, etc.) are properly protected with hard hats, steel-toe boots, gloves, etc., as required.

4. TEST PROCEDURES.

4.1 Selection of Method. Prior to conducting a Center of Gravity (CG) determination for a test item, consideration must be given to the selection of the proper procedure to be followed. The methods for determination of CG outlined in this Test Operations Procedure are discussed below. Whatever test procedure is selected, care and attention to detail must be followed during CG determination to eliminate possible execution errors.

a. Suspension Method. This method involves only the static suspension of the test item and no calculations are required to locate the CG. This method is based on the principle that, for a freely suspended mass, a vertical line through the point of suspension will pass through the

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center of gravity.

b. Reaction Method. This method is used when the equipment required to locate the CG of a test item by the suspension method is of inadequate capacity to handle the weight of the test item, or when the design and/or configuration of the test item are such that it cannot be suspended. This method is based on the principle that the sum of the moments about an axis of rotation is zero, so long as the body is in a static position.

c. Weight Method. This method is based on the principle of weight distribution and is most useful for vehicles with two axles although multiple axle and tracked configurations can be tested with this method. Longitudinal (x) and lateral (y) center of gravity locations can be determined with this method. Many transportability analyses only require a longitudinal (x) location.

d. Pendulum Timing Method. This method is based on the fact that the vertical center of gravity location can be calculated using the periods of oscillation measured when the test item is suspended and swung in a pendular motion at two different pendulum lengths. This method is best for vertical CG location and moments of inertia; other CG measurements may be impractical as they require awkward positioning of the test item. Sloshing fluids must be eliminated to ensure accuracy with this method.

e. Combined CG Method. This method is used when the configuration of several attached masses makes it impossible or impractical to obtain the combined CG by one of the above empirical methods. This method is based on the principle that the combined CG can be determined by taking moments about a known CG or common reference point.

4.2 Suspension Method (used for CG along the longitudinal and vertical axes).

a. Suspend the test vehicle by cables of unequal length fastened to lifting attachments at each end and hanging freely from the hook of a crane (Figures 1 & 2). The lengths of the cables determine the suspension angle; an angle of 45° is the optimal condition. The lifting eyes must be symmetrical to the vehicle's longitudinal and lateral centerlines for this method to be accurate. Care must be exercised to ensure a level lift perpendicular to the axis of the transit and crane hook. Additional shackles may be required in individual legs of the rigging to guarantee the planarity of the vehicle.

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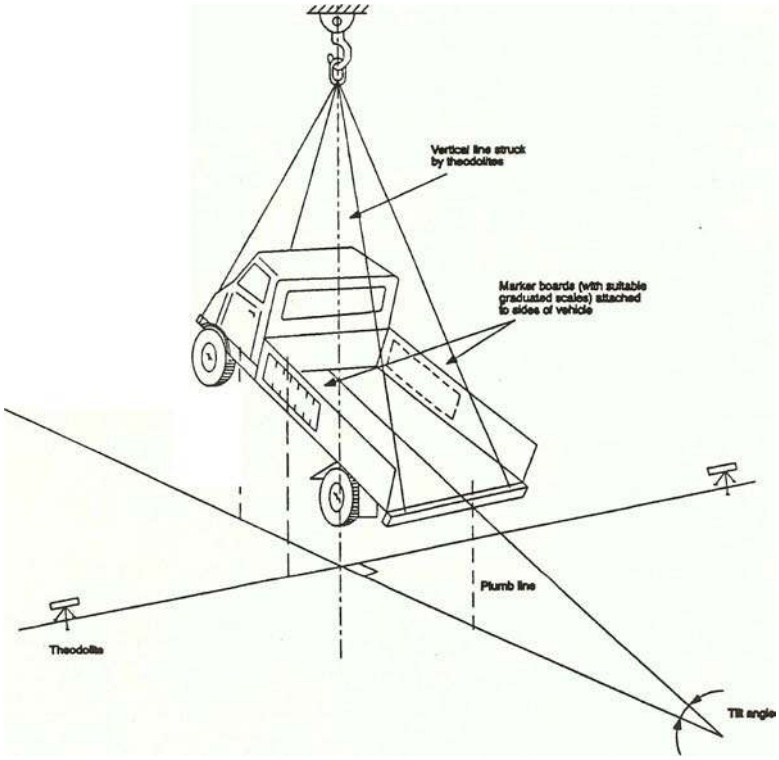


Figure1. Depiction of Suspension Method

b. Using a transit or theodolite, project a vertical line from the center of the load hook pivot and inscribe it on a marking board attached to the side of the vehicle; the inscribed line will be contained in a vertical plane which also contains the CG. Ensure that the vertical plane of the test vehicle longitudinal-axis is normal to the vertical plane described by the sighting scope (transit). Per SAE J874[9], if a scribing board is required it shall be at least 24 inches in height and 18-inches wide, rigidly constructed, and attached to the test item in a suitable position with a smooth face vertical and parallel to the longitudinal centerline of the vehicle.

c. Repeat the above operation with the suspension cables interchanged in order to locate the CG in the longitudinal and vertical directions (intersection of the two projected lines).

d. Inscribing the intersecting lines on the marking board projects the CG to a vertical plane at the side of the vehicle. Measure and record the distances between the point of intersection (CG) and horizontal and vertical planes drawn through a reference point on the vehicle body which is not affected by variations in suspension (e.g., tire pressure. suspension sag, etc.). Height of the CG reference point above ground level is recorded as an additional measurement. Operate the vehicle in the shop environment to allow the suspension to settle prior to making the height measurements.

e. Calculate the lateral location of the CG using the weight method described in paragraph 4.4. Locate the CG relative to the vehicle's longitudinal centerline.

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4.3 Reaction Method (used for CG along the longitudinal and vertical axes).

a. Position the vehicle (with brakes released) on a level surface and attach lifting cables, with force-measuring device (load cell) attached, to one end. Suspend the end of the vehicle by means of a crane so that it pivots about the axis of rotation (forward or rear axle); the tilt angle should be at least 25°. An alternate means for the pivot can be accomplished using a separate crane and rigging. If the vehicle design permits (such as an angular hull), a knife edge axis of rotation can be constructed from wooden timbers with a steel interface. Care should be exercised to ensure the crane lift is parallel to imaginary plane described by the crane rigging and load cell. After the desired angle has been reached, ensure that the vehicle is free of motion and the lifting force is being applied in a true vertical direction. Figure 2 shows a typical wheeled vehicle weight reaction test. Figure 3 shows the weight reaction method on a tracked vehicle.



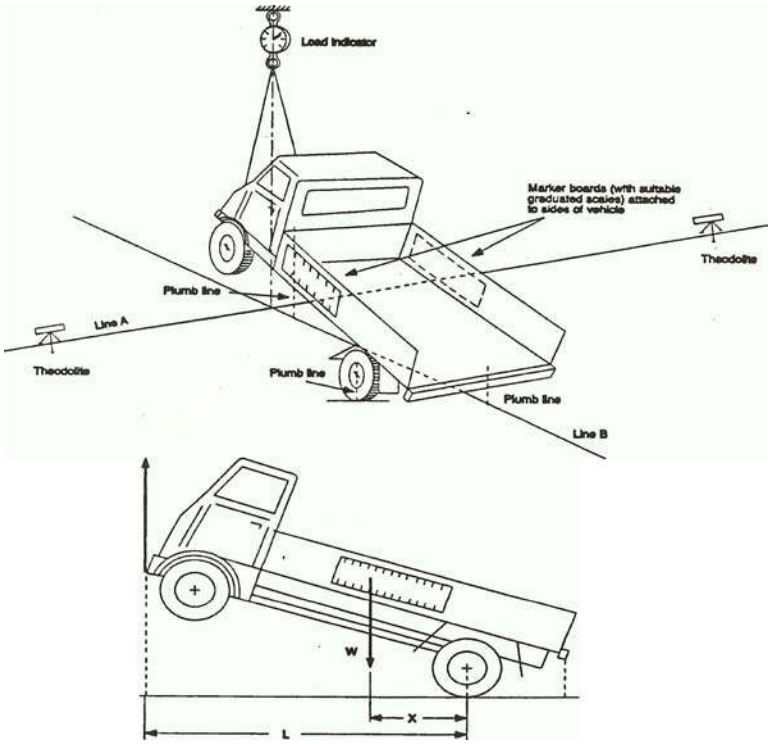
Figure 2. Weight Reaction Method (Wheeled Vehicles)



Figure 3. Weight Reaction Method (Tracked Vehicle)

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b. Determine the distance between the axis of rotation and the CG by the equation presented in Figure 4. Values for L, S, and W are obtained by measurement; the value for x is calculated as shown. Project a vertical line, by means of a plumb bob, transit, or theodolite, at the calculated distance (x) from the axis of rotation, and inscribe it on a marking board attached to the side of the vehicle.



$$x = SL/W$$

where:

x - Longitudinal distance of CG from axis of rotation.
S – Measured force in lifting cables.
L – Projected longitudinal distance between axis of rotation and center of lifting eye.
W - Vehicle test weight.

Figure 4. Weight Reaction Method

c. Reverse the suspension device and the axis of rotation so that the vehicle pivots about the axis of rotation (forward or rear axle) at its opposite end. Take necessary measurements as above, calculate the distance of the CG from the axis of rotation, and inscribe a second line on the marking board which intersects the first line. The point of intersection locates the CG in longitudinal and vertical directions. The longitudinal measurement should be referenced through

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the driving sprocket axle for a tracked vehicle. For wheeled vehicles the reference is generally through the rearmost non-steering axle. Transportability analyses require the longitudinal location to the most forward hard point on the vehicle (generally the front bumper). Axle spacing and front and rear overhang should be provided with the location giving the analyst the ability to move the reference as required.

d. Inscribing the intersecting lines on the scribing board projects the CG to a vertical plane at the side of the vehicle. Measure and record the distances between the point of intersection (CG) and horizontal and vertical planes drawn through a reference point on the vehicle body which is not affected by variations in suspension (e.g., tire pressure, suspension sag, etc.). Height of the CG reference point above ground level is recorded as an additional measurement. A hard contact shall be assumed, i.e. no grouser penetration in the case of a tracked machine. Additional physical characteristics are often required to fully characterize the CG location. If reference to the ground plane is used, axle centerline heights above the ground plane should be measured. Tire pressures and tire side wall deflection should also be measured to fully characterize the vehicle. Final height measurements should be taken only after operating the vehicle in the shop environment to allow the suspension to settle during the lifting process.

e. Calculate the lateral location of the CG using the weight method described in paragraph 4.4. Locate the CG relative to the vehicle longitudinal centerline.



Figure 5. Weight Reaction Method Using a Platform Scale in lieu of a Load Cell.

NOTE: The same result can be achieved, through similar calculations, by measuring the weight reaction under the axis of rotation (utilizing a platform scale) instead of the vertical lifting force at the opposite end. (Figure 5)

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4.4 Weight Method (used for CG along all three axes).

a. Measure weight reactions P_1 and P_2 at each vehicle axle by means of platform scales. Also measure distance e between axles, i.e., wheelbase. Calculate a and/or b using the formula in Figure 6. Care should be taken when using individual wheel scales. A suitable location is required. The location should be level to within 0.1% to minimize scale planarity errors.

b. Place the front (or rear) wheels of the vehicle on a raised platform (step) and the rear (or front) wheels on a platform scale. Knowing this weight reaction and the total vehicle weight, calculate the weight reaction at the opposite set of wheels. The ratio between the step height and the vehicle wheelbase should be no less than 0.25. This will equate to a minimum angular change of approximately 15 degrees[10].

c. Determine the vertical CG location by the equation shown in lower portion of Figure 7. Values for r, W, Y_2, and m are obtained by measurement; values for Y_1 and H are obtained by calculation.

d. Calculate the lateral location of the CG using the method presented in Figure 8. Locate the CG relative to the vehicle longitudinal centerline. The measurement of wheel track, between centers of tire contact, can be complicated by the existence of camber and toe-in. It is customary to measure wheel track above the road plane at the tire tread centerline at front and rear. These dimensions should be averaged to correct for toe-in[11].

$$b = \frac{P_1 e}{W}$$

$$a = \frac{P_2 e}{W}$$

where:

b = Distance of CG from forward axle.
a = Distance of CG from rear axle.
P_1 = Forward axle weight reaction.
P_2 = Rear axle weight reaction.
e = Wheelbase.
W = Total vehicle weight

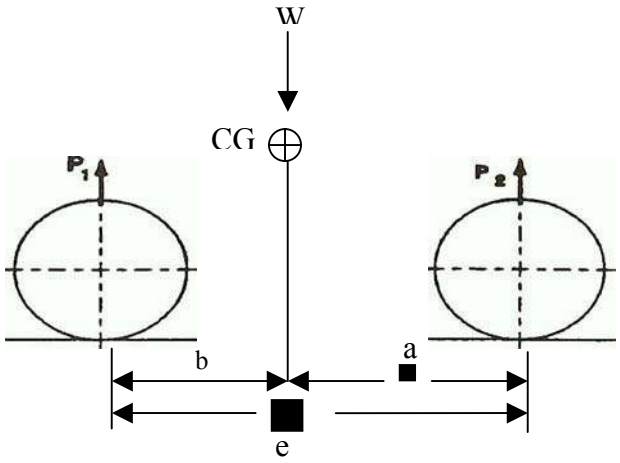
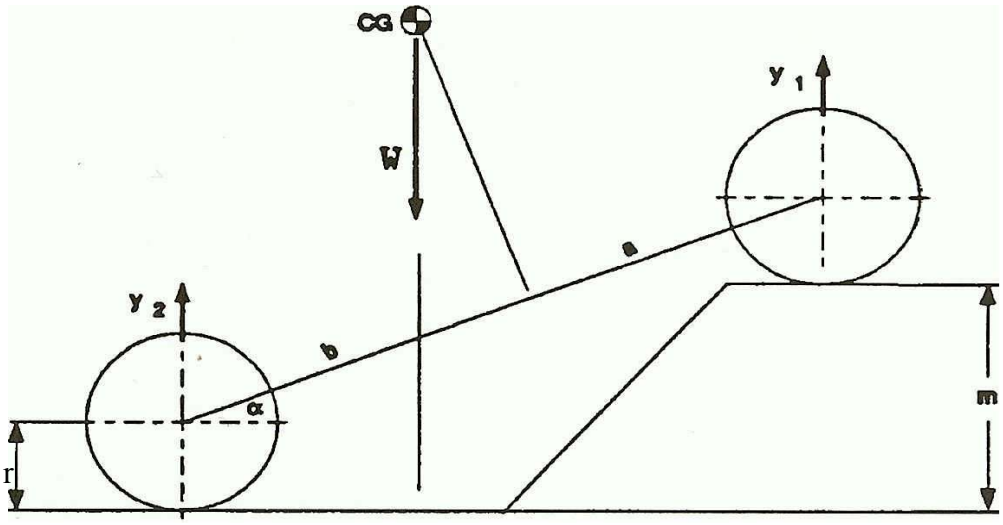


Figure 6. Weight Method.

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Vehicle with one Axle Raised

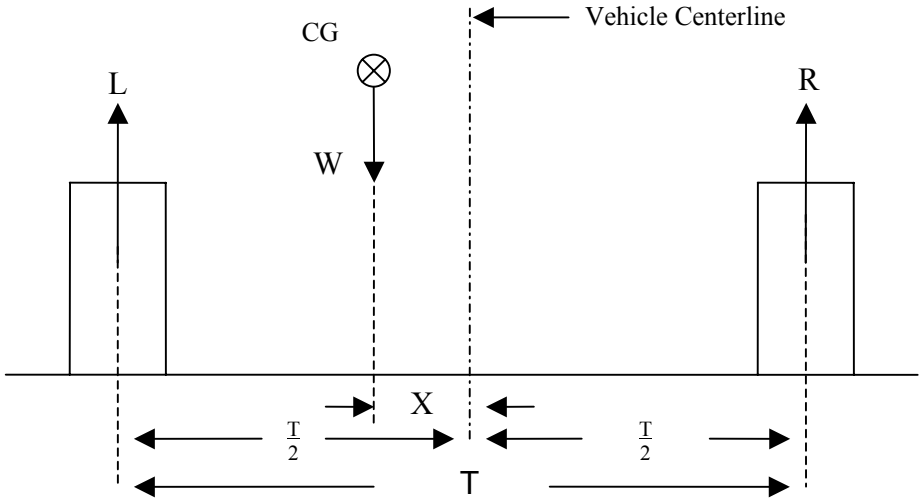
$$H = \frac{Y_2\, e - Wa}{Wm} \sqrt{e^2 - m^2} + r$$

where:

H = Height of CG above ground level.
Yz = Weight reaction (platform scale reading).
e = Wheelbase.
W = Total vehicle weight.
a = Longitudinal distance between CG and center of raised axle (when level).
m = Step height.
r = Radius of wheel and tire combination.

Figure 7. Step Method.

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Taking moments about left side,

$$X = \frac{T\ (W - 2R)}{2W}$$

or, taking moments about right side,

$$X = \frac{T\ (2L - W)}{2W}$$

where:

X = Distance of CG from vehicle centerline.
T = Tread.
W = Total vehicle weight.
L = Vehicle left side weight reaction.
R = Vehicle right side weight reaction.

Figure 8. Calculation of Lateral CG Location.

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4.5 Pendulum Timing Method (used when vehicle moments of inertia are required).

a. Suspend each end of the vehicle from a freely rotating hub using rigid links. The connections between the links and the hubs and the vehicle must not allow any relative motion.

b. Measure the vertical distance from the hub centerlines to any specific vehicle horizontal reference plane. This distance will be considered the link length.

c. Set the suspended mass into pendular motion, allowing a maximum 5° swing from the rest position.

d. Time the period of the pendular motion to an accuracy of 1 milli-second, averaging the value obtained over 10 periods.

e. Shorten or lengthen the rigid links by approximately 30 to 60 cm. Repeat steps b, c, and d.

f. Determine the vertical CG location using the equation presented in Figure 9.

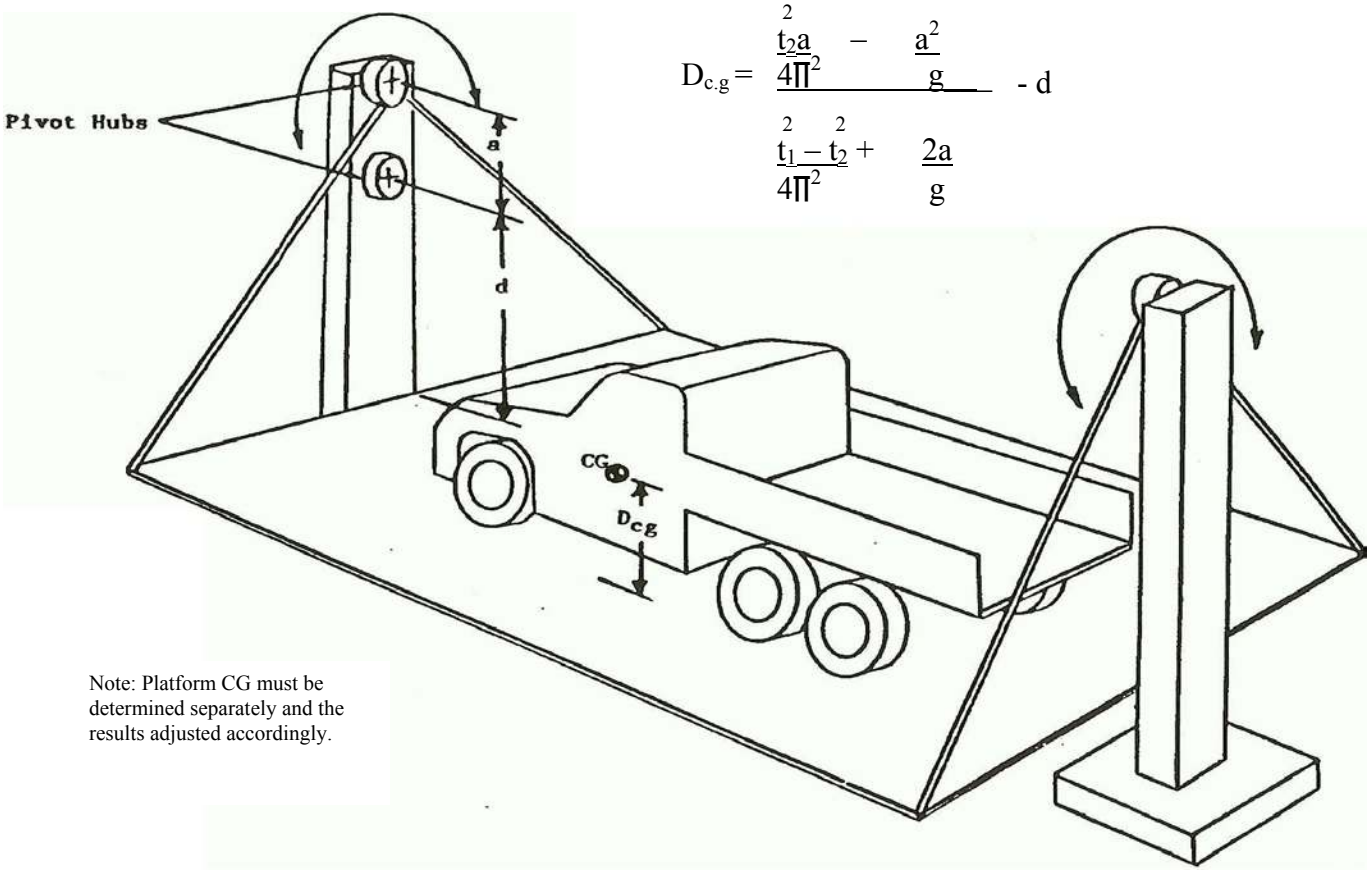
g. With the vehicle on the ground at its proper ride height, measure the vertical distance between the reference plane and ground level; with this distance and the distance between the reference plane and the CG (D_{cg}), the vertical location of the CG can be described. (In the example shown in Figure 10, the horizontal reference plane is at ground level.)

h. This method of CG determination can also be used to determine lateral and longitudinal CG locations. However, determination of lateral CG location requires that the vehicle be suspended with one side down (rolled 90° from horizontal), and determination of longitudinal CG location requires that the test item be suspended with one end down (pitched 90° from horizontal). Figure 10 shows a vehicle configured for a roll moment of inertia test.

i. The pendulum timing method of determining CG location also yields the measurements necessary to determine the moments of inertia of the test vehicle.

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$$D_{c.g} = \frac{\frac{t_2^2 a}{4\pi^2} - \frac{a^2}{g}}{\frac{t_1^2 - t_2^2}{4\pi^2} + \frac{2a}{g}} - d$$



Note: Platform CG must be determined separately and the results adjusted accordingly.

Figure 9. Pendulum Timing Method.

D_{cg} = Vertical distance between chosen horizontal reference plane and vehicle CG.
t_1 = Oscillation period for short-link pendulum.
t_2 = Oscillation period for long-link pendulum.
a = Difference between short- and long-pendulum link lengths.
g = Acceleration of gravity.
d = Short-link length (reference plane in this example is at ground level).

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Figure 10. HMMWV Configured for Measurement of Roll Moment of Inertia (Pendulum Method)

4.6 Combined CG Calculations.

a. Two masses. The combined CG of two masses for which the CG has previously been determined individually (e.g., a prime mover carrying a shelter or another item of equipment) can be determined by taking moments about a known CG as shown in Figure 11.

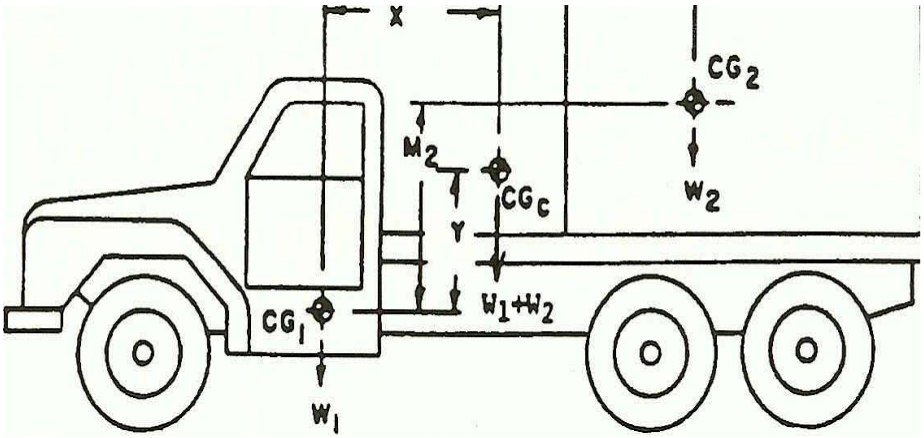
b. Three or more masses. To determine the combined CG of three or more masses for which the CG locations have been previously determined, use the same approach as below (fig. 11). The equation for locating the combined CG of three masses in the horizontal direction would then be as follows:

$$W_3 \ L_3 + W_2 \ L_2 = X \ (W_1 + W_2 + W_3)$$

or

$$X = \frac{W_3 \ L_3 + W_2 \ L_2}{W_2 + W_2 + W_3}$$

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CG_1 = Prime mover center of gravity.
CG_z = Payload center of gravity.
CG_c = Combined center of gravity of prime mover with payload.

To locate the longitudinal component of the combined CG, moments are again taken about CG_1:

$$W_2L_2 = X(W_1 + W_2)$$

or

$$X = \frac{W_2L_2}{(W_1 + W_2)}$$

where:

W_2 = Weight of payload.
L_2 = Longitudinal distance between CGl and CGz.
X = Longitudinal distance between CGl and CGc.
W_1 = Weight of prime mover.

To locate the vertical component of the combined CG, moments are again taken about CG_1:

$$Y = \frac{W_2M_2}{(W_1 + W_2)}$$

or:

$$Y = W_2M_2 = Y(W_1 + W_2)$$

where:

M_2 = Vertical distance between CG_l and CG_2.
Y = Vertical distance between CG_l and CG_c.

Location of the lateral component is determined in a similar manner.

Figure 11. Method of Calculating Combined Center of Gravity.

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5. DATA REOUIRED.

5.1 All Methods.

a. Vertical, lateral, and longitudinal center of gravity coordinates (±25 mm) with respect to a reference point on the vehicle. The vehicle axis system is defined by SAE J670e[12] as a right hand orthogonal axis system fixed in the vehicle. x-axis points forward and is in the longitudinal plane of symmetry. y-axis points to the driver's right and the z-axis points down.

b. Customary center of gravity reference locations for wheeled vehicles are the following:

x-axis: Forward of the rear axle centerline presuming the rear the rear axle is a non steering axle. If the rear axle is a steering axle care must be taken to insure steering affects are not added into the measurement. For transportability studies and aircraft loading the longitudinal location is required to be referenced from the front bumper. This can be calculated as long as the front overhang and wheelbase measurements are included. NATO Reference Mobility Model (NRMM)[13] uses the rear axle centerline and the ground plane as the reference points for the longitudinal and vertical c.g. locations, respectively.

y-axis: Left or right of the vehicles longitudinal centerline.

z-axis: Above the ground plane. A secondary measurement should be provided to include the axle centerline heights above the ground plane.

c. Customary center of gravity reference locations for tracked vehicles are the following:

x-axis: Forward or rearward of the sprocket centerline. For transportability studies and aircraft loading the longitudinal location is required to be referenced from the front glacis.

y-axis: Left or right of the vehicles longitudinal centerline.

z-axis: Above the ground plane. A secondary measurement should be provided to include the sprocket centerline height above the ground plane.

d. Vehicle loading conditions.

e. Payload reference locations.

5.2 Reaction Method.

a. Total vehicle weight.

b. Lifting force.

c. Longitudinal distance between axis of rotation and vertical line of lifting force.

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5.3 Weight Method.

a. Total vehicle weight.

b. Left and right side weight distribution.

c. Axle weight distribution.

d. Tread measurement.

e. Longitudinal distances between axle centerlines.

f. Weight on one axle with opposite end of vehicle on step.

g. Wheelbase.

h. Tire radius.

i. Step height.

5.4 Pendulum Timing Method.

a. Link lengths.

b. Periods of pendulum oscillation.

6. DATA PRESENTATION.

a. Locate information on photograph or sketch.

(1) Identify and locate reference point in all three axes.

(2) Locate CG in longitudinal and vertical axes with respect to the reference point.

(3) Locate CG in lateral axis relative to vehicle longitudinal centerline.

b. State method used and rationale for selection of method.

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APPENDIX A. REFERENCES

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Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: Test Business Management Division (CSTE-DTC-TM-B), US Army Developmental Test Command, 314 Longs Corner Road Aberdeen Proving Ground, MD 21005-5055. Technical information may be obtained from the preparing activity: Automotive Instrumentation Division (CSTE-DTC-AT-AD-I), US Army Aberdeen Test Center, 400 Colleran Road, Aberdeen Proving Ground, MD 21005-5059. Additional copies are available from the Defense Technical Information Center, 8725 John J. Kingman Rd., STE 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

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